

Conformal Correlation Counterexamples

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Abstract

Conformal prediction has recently been proposed for the diagnosis of classifiers and the selection of federated clients, with the prediction sets themselves treated as data from which class similarity, confusion and client quality are read off. We give a specific caution in the form of a no-go theorem and a one-bit calculation. Over the class of joint laws constrained only by $\hat{Y} \in C(X)$ whenever $C(X) \neq \emptyset$, the law of the prediction sets does not identify the confusion matrix once a set of size at least two occurs, and the corresponding failure inside randomized adaptive prediction sets is exhibited by construction. We illustrate with the conformal correlation matrix of Perlo et al. (2026), the phi coefficient between set membership indicators, a natural construction that is likely to be reinvented. Two classifiers built inside randomized adaptive prediction sets, exactly as Romano, Sesia and Candès specify them, produce the same matrix while confusing every class in opposite directions. Seven further hand calculations give a zero entry beside 99% directional confusion, an entry of one beside one in a million co-occurrence, and an entry of ± 0.64 from pooling two clients with zero correlation on each. For classes that never share a set the entry is a function of the inclusion rates alone, and for a perfect classifier with equal top probabilities inside randomized adaptive prediction sets the matrix reveals a client’s class proportions exactly, which undoes the privacy claim made for it.

1 The statistic

There are H classes. A classifier maps an input X to a probability vector $f(X)$, with $f_h(X)$ the probability it assigns to class h . Split conformal prediction (Vovk et al., 2005; Lei et al., 2018) turns the probability vector into a prediction set $C(X) \subseteq \{1, \dots, H\}$. The true label is Y and the class the classifier ranks first is $\hat{Y}$.

The authors build their sets with the adaptive prediction sets (APS) of Romano et al. (2020). APS assigns a score to each label at each input. Sort the probabilities at x in decreasing order, $f_{(1)}(x) \geq \dots \geq f_{(H)}(x)$, with ties broken at random and zero-probability labels discarded, as Romano et al. (2020) suggest for their oracle, write $\text{rk}(x, y)$ for the rank of class y in that order, and write $S_k(x) = \sum_{l \leq k} f_{(l)}(x)$ for the probability mass through rank k . The score of label y at input x is

$$s(x, y, u) = S_{\text{rk}(x, y)}(x) - u f_y(x), \quad u \sim \text{Unif}(0, 1), \quad (1)$$

the mass at and above y in the ranking, less a uniform share of the mass of y itself.

The score is turned into a set by calibration. Given m labeled calibration pairs (x_i, y_i) with independent uniforms u_i , let $k = \lceil (m+1)(1-\alpha) \rceil$ and let τ be the k -th smallest of the calibration scores $s(x_i, y_i, u_i)$, the sample quantile of Lei et al. (2018).¹ If $\alpha < 1/(m+1)$ then $k = m+1$ and

¹Algorithm 1 of Romano et al. (2020) prints “largest” where the proof of their Theorem 1 uses the $(1-\alpha)$ empirical quantile, which is the k -th smallest, and their set drops the boundary class at exact equality of score and threshold where (2) keeps it. Neither difference matters here, since test scores have a density.

no such score exists, in which case $\tau = +\infty$ and the set is every class. We assume $\alpha \geq 1/(m+1)$ throughout, so that $k \leq m$. The prediction set at a new input x , with a fresh uniform U , is

$$C(x) = \{y : s(x, y, U) \leq \tau\}. \quad (2)$$

In words, the set is the top-ranked classes down to the first rank at which the cumulative mass reaches τ , and the class at that boundary rank is kept or dropped according to U . When the boundary rank is the first and the class is dropped, the set is empty.

Two coverage bounds hold. Under exchangeability of the calibration pairs and the test pair, $\mathbb{P}(Y \in C(X)) \geq 1 - \alpha$. When the scores are almost surely distinct, also $\mathbb{P}(Y \in C(X)) \leq 1 - \alpha + 1/(m+1)$ (Romano et al., 2020, Theorem 1). Examples 1 and 9 use the randomized variant exactly as defined here. Theorem 1 uses only that the top-ranked class is in every nonempty set, and the remark on α in Section 4 uses only that the sets are nested in α . Nothing else below depends on APS, beyond the fact that $C(X)$ is a set of classes.

The matrix is built from membership indicators. For each class h define

$$z_h = \mathbf{1}\{h \in C(X)\},$$

and write $p_h = \mathbb{E}[z_h]$ for the inclusion rate of class h and $q_{ij} = \mathbb{E}[z_i z_j]$ for the co-occurrence rate of classes i and j . The conformal correlation matrix (CCM) of Perlo et al. (2026) has entries

$$\rho_{ij} = \frac{\mathbb{E}[z_i z_j] - \mathbb{E}[z_i] \mathbb{E}[z_j]}{\sqrt{\text{Var}(z_i) \text{Var}(z_j)}} = \frac{q_{ij} - p_i p_j}{\sqrt{p_i(1-p_i) p_j(1-p_j)}}. \quad (3)$$

This is the Pearson correlation of two Bernoulli variables, also called the phi coefficient (Yule, 1912). In practice the expectations are sample averages over a test set.

The entry is defined only when $0 < p_i < 1$ and $0 < p_j < 1$, since otherwise a variance is zero. The authors give no convention for that case in their Section 4, where the matrix is defined. We exclude such classes from the matrix.

Nothing below depends on the name or on the paper. The results concern any correlation between membership indicators of prediction sets, however the sets were built and whatever the statistic is called. The paper is the instance in print, and the move it makes, correlating the indicators and reading the result as class similarity, is cheap enough that it will be made again. Raw co-occurrence counts of classes in prediction sets already appear as a visualization in Garcia-Ceja et al. (2023), as counts and as column-normalized proportions with no correlation taken, and the paper does not cite that work. The correlation is the paper’s addition, and it is where the trouble starts.

The authors attach three readings to the entries. Their Remark 1 says that intrinsically similar classes “might be expected to exhibit high positive correlation”. Their Remark 2 says that pairs of dissimilar classes “are expected to display strong negative correlation”. Their Remark 3 says that classes “unrelated or independent with respect to the classifier representation are expected to yield correlation coefficients close to zero”.

On the strength of these readings they draw three conclusions. A positive entry beside a negative one means the classifier “is much more likely to misclassify” the focus class as the first partner than as the second, in their Benefit 1 and Figure 2. The matrix “contains a superset of the information conveyed by the confusion matrix”. And the matrix “captures the likelihood of class pairs co-occurring within conformal prediction sets”. Each of the three is false, and the examples below show it.

2 The matrix depends only on the law of the sets

Observation 1. The conformal correlation matrix is a functional of the law of the prediction set $C(X)$ alone. Two situations with the same law of $C(X)$ have the same matrix, whatever the joint law of $(C(X), Y, \hat{Y})$.

Proof. Every quantity in (3) is an expectation of a function of the vector $(z_1, \dots, z_H)$, and that vector is a function of $C(X)$. $\square$

Confusion is a different object. The confusion matrix has entries $\mathbb{P}(Y = a, \hat{Y} = b)$, which depend on the joint law of the true label and the ranked prediction. Hold the law of $C(X)$ fixed and change Y or $\hat{Y}$, and by Observation 1 the matrix cannot notice.

The observation also disposes of the claim that the entries “incorporate the entire distribution of predicted probabilities”. At a fixed α , any change to the probability vectors that leaves the sets unchanged is invisible to the matrix. The sets are a function of the probabilities, the uniform U and one threshold, and the matrix sees the sets.

The marginal coverage guarantee ties $C(X)$ to Y through one scalar constraint, and a construction may impose more. Under exchangeability of calibration and test data, split conformal prediction guarantees

$$\mathbb{P}(z_Y = 1) = \sum_y \mathbb{P}(Y = y, z_y = 1) \geq 1 - \alpha. \quad (4)$$

This is one number. It constrains the sum on the right and nothing else.

In particular, coverage does not in general determine a co-occurrence rate q_{ij} . That rate mixes the true class with a wrong class, two wrong classes, and different true classes on different inputs, and coverage never justifies reading it as confusion. One special case is exact. With only two classes and coverage one the set is never empty, so $0 = \mathbb{P}(C(X) = \emptyset) = 1 - p_A - p_B + q_{AB}$ and $q_{AB} = p_A + p_B - 1$ is forced. With three or more classes nothing of the kind holds, as the next lemma shows for a single entry.

Lemma 1. *With at least three classes, for every $\rho \in [-1, 1]$ there is a joint law of $(Y, \hat{Y}, C(X))$ with coverage one and accuracy one whose matrix has an off-diagonal entry equal to ρ .*

Proof. Let the true label always be T , always in the set and always ranked first, so coverage and accuracy are one. Since $z_T \equiv 1$ the entries involving T are undefined, and the claim concerns ρ_{AB} . Let two other labels A and B be added to the set according to

$$\begin{aligned} \mathbb{P}(z_A = 0, z_B = 0) &= \mathbb{P}(z_A = 1, z_B = 1) = \frac{1 + \rho}{4}, \\ \mathbb{P}(z_A = 1, z_B = 0) &= \mathbb{P}(z_A = 0, z_B = 1) = \frac{1 - \rho}{4}. \end{aligned}$$

Then $p_A = p_B = \frac{1+\rho}{4} + \frac{1-\rho}{4} = \frac{1}{2}$, so each variance is $\frac{1}{4}$, and $q_{AB} = \frac{1+\rho}{4}$. Substituting into (3),

$$\rho_{AB} = \frac{(1 + \rho)/4 - 1/4}{1/4} = \rho. \quad \square$$

In the construction Y is constant, so every value on the correlation scale is available with no classification problem at all.

The reasoning of Observation 1, that the matrix cannot see $\hat{Y}$, gives a no-go result that covers the conformal correlation matrix and any other statistic computed from the prediction sets.

Theorem 1 (No statistic of the sets determines the confusion matrix). *Call a joint law of $(Y, \hat{Y}, C(X))$ admissible if $\hat{Y} \in C(X)$ whenever the set is nonempty, as under APS. Fix a joint law of $(Y, C(X))$ under which some set of size at least two occurs with positive probability. Then there are two admissible joint laws of $(Y, \hat{Y}, C(X))$, each with the fixed joint law of $(Y, C(X))$, hence the same law of sets, the same class proportions, the same coverage and the same value of every functional g of the law of sets, whose confusion matrices differ.*

Proof. Let B be a set with $|B| \geq 2$ and $\mathbb{P}(C(X) = B) = \delta > 0$, and let $b \neq b'$ be two of its elements. Extend the fixed joint law of $(Y, C(X))$ to an admissible joint law of $(Y, \hat{Y}, C(X))$, for instance by setting $\hat{Y}$ to the smallest element of $C(X)$ when the set is nonempty and to a fixed class otherwise. Define a second joint law that agrees with this one except on the event $\{C(X) = B\}$, where it sets $\hat{Y} = b$, and a third that agrees with it except on that event, where it sets $\hat{Y} = b'$. The three laws have the same joint law of $(Y, C(X))$, hence the same law of sets, the same class proportions, the same coverage $\mathbb{P}(Y \in C(X))$, and the same value of every functional g of the law of sets. Pick a class a with $\mathbb{P}(Y = a, C(X) = B) = \delta_a > 0$. The second law has confusion entry $\mathbb{P}(Y = a, \hat{Y} = b)$ larger by δ_a than the third, and entry $\mathbb{P}(Y = a, \hat{Y} = b')$ smaller by δ_a . So the second and third confusion matrices differ. $\square$

The conclusion is stronger than the title. Even the entire joint law of $(Y, C(X))$ does not determine the confusion matrix, because both constructed laws preserve it. What is lost is the information about the rankings, not only the information about the true label. Figure 1 places the two losses on the chain of coarsenings that leads from the data to the matrix.

Two limits on the theorem. First, its hypothesis is sharp in the following sense. If every set is a singleton then $\hat{Y}$ is its unique element, the confusion matrix is $\mathbb{P}(Y = a, C(X) = \{b\})$, and the joint law of $(Y, C(X))$ determines it, whatever the number of classes or the coverage. With coverage one it is the diagonal matrix of the class proportions. If instead the empty set occurs with positive probability, the ranked prediction is unconstrained on that event and the argument of the proof applies there. So over admissible laws the joint law of $(Y, C(X))$ determines the confusion matrix exactly when the sets are almost surely singletons. Second, the construction changes $\hat{Y}$ while keeping $C(X)$, and that need not be realizable by an APS classifier with its calibration, so the theorem does not establish APS realizability for every prescribed set law. Example 1 supplies a pair that is realized by two randomized APS classifiers.

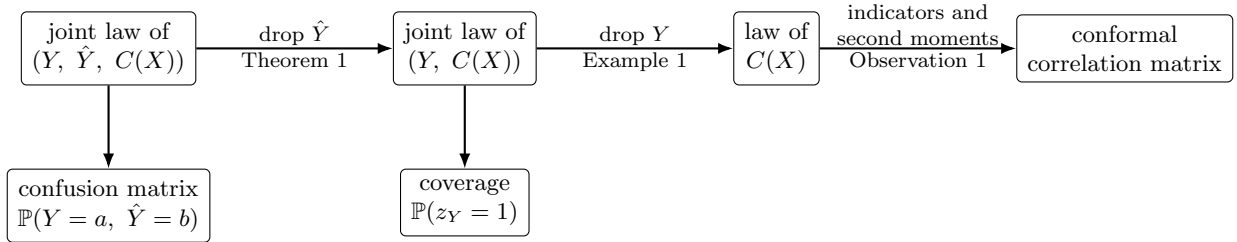


Figure 1: The chain of coarsenings from the data to the matrix. The confusion matrix is a function of the first object. Theorem 1 shows it can be lost at the first step, where the ranked prediction is dropped. Example 1 shows a loss of one bit at the second step, where the true label is dropped, by a different construction: in that mixture the ranked prediction is a function of the label and the set, so the first step loses nothing there and the second loses the bit. The two results show failures at different arrows. The coverage guarantee constrains one scalar of the second object. The matrix is a function of the third.

The examples that follow are more concrete. Together they show that the sign and magnitude of an entry cannot be used as evidence of similarity, dissimilarity, independence or directional confusion.

3 Eight examples

Each example needs only the inclusion rates p_i and the co-occurrence rates q_{ij} , substituted into (3).

Examples 2 through 8 prescribe a law of prediction sets directly. By Observation 1 the law of the sets is all the matrix depends on, so a law is all an example needs to specify.

In those examples coverage is one, meaning that every set contains the true class. Any marginal coverage lower bound is then met. We do not claim that randomized APS produces these laws. Under exchangeability, when $k \leq m$ and the scores are almost surely distinct, it cannot produce coverage one, because its coverage is then exactly $k/(m+1) < 1$, the rank of the test score among the $m+1$ scores being uniform (Lei et al., 2018; Romano et al., 2020).

Therefore Examples 2 through 8 concern laws of prediction sets that meet the coverage bound, and show that the readings fail over that class. A claim restricted to sets produced by APS needs a counterexample built inside APS. Example 1 is that counterexample, which is why it comes first. The remaining examples are ordered with the simplest refutation and the most absurd result first.

In Examples 4, 6 and 8 a label T is in every set. Its own entries are undefined under the convention of Section 1, and the claims concern ρ_{AB} only.

Example 1 (Opposite confusions, the same matrix, within randomized APS). Three classes with equal frequency. Two classifiers assign the probabilities (0.6, 0.3, 0.1) to the classes in the orders shown, so the true class is always ranked second.

True class	Model 1 order	Model 2 order
A	$C > A > B$	$B > A > C$
B	$A > B > C$	$C > B > A$
C	$B > C > A$	$A > C > B$

The two models get the same threshold. By (1) the score of the true label is $0.9 - 0.3u$ under both models, uniform on (0.6, 0.9). Calibrate both on the same inputs with the same uniforms, with $k \leq m$ as assumed in Section 1, and both have the same threshold $\tau \in (0.6, 0.9)$. Write $r = (\tau - 0.6)/0.3$.

The two models produce sets with the same law. At a test input the top-ranked class has score $0.6 - 0.6U \leq \tau$ and is always included. The second has score $0.9 - 0.3U$, which is at most τ exactly when $U \geq (0.9 - \tau)/0.3$, an event of probability r . The third has score $1 - 0.1U \geq 0.9 > \tau$ and is never included. Under either model, therefore, each singleton $\{A\}$, $\{B\}$, $\{C\}$ occurs with probability $(1 - r)/3$ and each pair $\{A, B\}$, $\{B, C\}$, $\{C, A\}$ with probability $r/3$. The same law of sets gives the same matrix. For the record, $p_i = (1 + r)/3$ and $q_{ij} = r/3$ for every class and pair, so every off-diagonal entry is

$$\rho_{ij} = \frac{r/3 - (1 + r)^2/9}{(1 + r)(2 - r)/9} = -\frac{r^2 - r + 1}{(1 + r)(2 - r)},$$

the same for both models and negative for every $r \in (0, 1)$.

Coverage is the same for both models too. Conditional on the calibration sample, coverage equals r exactly. Since τ is the k -th smallest of m uniforms on (0.6, 0.9), the variable r is the k -th smallest of m uniforms on (0, 1), which has the Beta($k, m + 1 - k$) law with mean $k/(m + 1) \geq 1 - \alpha$.

So on a given calibration sample r can fall below $1 - \alpha$, and its average over calibration samples meets the marginal guarantee, as in Theorem 1 of Romano et al. (2020). The equality of the two set laws holds for every realized τ .

The two models confuse classes in opposite directions. Model 1 labels every A as C , every B as A and every C as B . Model 2 runs the cycle the other way. Their confusion matrices are transposes of each other, both with accuracy zero. The pair survives the deterministic variant too: with $u = 0$ every calibration score is 0.9, so $\tau = 0.9$, the second-ranked class is always included, every set is the pair of the top-ranked and the true class, and the two models again share the law of sets with opposite confusions.

The authors write that the matrix “contains a superset of the information conveyed by the confusion matrix”. Here the sets are produced by randomized APS exactly as Romano et al. (2020) specify, the matrix is the same for both models, and the confusion matrices differ. The matrix does not contain the confusion matrix.

The loss can be measured. Fix the threshold and choose between the two models by a fair coin D . The set law does not depend on D , so $I(D; C(X)) = 0$. But when $Y = A$, model 1 produces $\{C\}$ or $\{A, C\}$ and model 2 produces $\{B\}$ or $\{A, B\}$, and likewise for the other two classes, so the set together with its true label identifies D and $I(D; Y | C(X)) = \log 2$. The sets alone carry no information about the direction of confusion. The sets with their labels carry one full bit. Put as a loss, the optimal logarithmic loss for predicting D rises from zero using $(Y, C(X))$ to $\log 2$ using $C(X)$ alone. The matrix is computed from the sets alone.

Remark. The bit is a diagnostic analogue of the information gap of Cotton (2026), where pooling a conditional residual law into one marginal law costs exactly the mutual information between residual and input under the logarithmic score. There the pooled object is the residual law and the cost is a forecasting regret. Here the pooled object is the joint law of set and label, reduced to the law of the set, and the cost is the bit above. The ranked prediction is a function of set and label in this mixture, so including it changes nothing.

Example 2 (A perfect classifier and a perfectly wrong one share a matrix). This example makes the same point without the procedure, and with the accuracy gap as wide as it can be. Three classes A, B, C with equal frequency, and sets that always contain the true class and one neighbor.

True class	Prediction set	Good model ranks first	Bad model ranks first
A	$\{A, B\}$	A	B
B	$\{B, C\}$	B	C
C	$\{C, A\}$	C	A

Each label is in two of the three sets, so every $p_i = 2/3$ and every variance is $2/9$. Each pair shares exactly one of the three sets, so every $q_{ij} = 1/3$. Every off-diagonal entry is

$$\rho_{ij} = \frac{1/3 - (2/3)(2/3)}{2/9} = -\frac{1}{2}.$$

The good model has accuracy one and the bad model has accuracy zero. The labels and the sets are identical for both, so the matrix is identical for both. The confusion matrix $\mathbb{P}(Y = a, \hat{Y} = b)$ of the good model has all its mass on the diagonal and that of the bad model has none there. The matrix does not contain “a superset of the information conveyed by the confusion matrix”.

Example 3 (Zero correlation with 99% directional confusion). Three classes and a small number ε , say 0.01. Inputs arrive in four kinds.

Probability	True class	Prediction set	Ranked first
$(1 - \varepsilon)/2$	A	$\{A, B\}$	B
$\varepsilon/2$	A	$\{A\}$	A
$(1 - \varepsilon)/2$	B	$\{B\}$	B
$\varepsilon/2$	C	$\{C\}$	C

Class A is in the sets of the first two kinds, so $p_A = (1 - \varepsilon)/2 + \varepsilon/2 = 1/2$. Class B is in the sets of the first and third kinds, so $p_B = 1 - \varepsilon$. They appear together only in the first kind, so $q_{AB} = (1 - \varepsilon)/2 = p_A p_B$. The numerator of (3) is zero, so

$$\rho_{AB} = 0.$$

For Bernoulli indicators that are not constant, zero correlation is independence. So z_A and z_B are independent here. Yet among inputs whose true class is A , the classifier ranks B first with probability $1 - \varepsilon$, which is 99% at $\varepsilon = 0.01$.

The authors' Remark 3 says that unrelated classes give an entry near zero. A reader of a heat map uses the converse, that an entry near zero means the classes are unrelated. The converse fails. Independence of set membership does not imply absence of classification confusion. The example also bears on their Remark 1 in its stated direction: A and B are as similar as two classes can be in the classifier's own representation, since it confuses them on 99% of the A inputs, and the entry is zero.

Example 4 (Perfect correlation with one in a million co-occurrence). Every input has the same true class T , which is in every set and ranked first, and A and B never occur as true labels. The labels A and B are added to the set together with probability ε , and otherwise neither is added. Then $z_A = z_B$ on every input, so

$$\rho_{AB} = 1 \quad \text{for every } 0 < \varepsilon < 1.$$

The probability that A and B appear together in a set is $q_{AB} = \varepsilon$, which may be 0.4, or 0.01, or one in a million. The authors write in their abstract that the matrix “captures the likelihood of class pairs co-occurring”. The likelihood is ε . The entry is one regardless.

Example 5 (Every pair positive from difficulty alone). Three classes with equal frequency, and difficulty independent of the true class. Half the inputs are hard and receive the set of all three classes. The other half are easy and receive the singleton set of their true class. The true class is ranked first in both cases, so accuracy is one.

Each label is in every hard set and in a third of the easy sets, so

$$p_i = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot \frac{1}{3} = \frac{2}{3}, \quad p_i(1 - p_i) = \frac{2}{9}.$$

Two labels appear together only in hard sets, so $q_{ij} = 1/2$ for every pair, and

$$\rho_{ij} = \frac{1/2 - (2/3)(2/3)}{2/9} = \frac{1}{4}.$$

Every pair is positively correlated and the classifier is perfect. The correlation has one cause, whether the input was hard, and owes nothing to any relation between the pair. The authors read a positive entry as confusion between the pair, in their Benefit 1, which is the converse of their Remark 1. Here every entry is positive, every entry is equal, and no pair plays any role.

Example 6 (Pooling two clients creates a strong correlation from nothing). Every input has the same true class T , which is in every set and ranked first, and A and B never occur as true labels. On client 1 the labels A and B are each added to the set independently with probability 0.9. On client 2 they are each added independently with probability 0.1. The clients contribute equally to the pooled data.

Within each client z_A and z_B are independent, so $\rho_{AB} = 0$ on each. Pooled, $p_A = p_B = (0.9 + 0.1)/2 = 1/2$, each variance is $1/4$, and

$$q_{AB} = \frac{1}{2}(0.9 \times 0.9) + \frac{1}{2}(0.1 \times 0.1) = 0.41, \quad \rho_{AB} = \frac{0.41 - 1/4}{1/4} = 0.64.$$

The sign can be reversed. Let client 1 add A with probability 0.9 and B with probability 0.1, and let client 2 do the opposite. Then $q_{AB} = \frac{1}{2}(0.09) + \frac{1}{2}(0.09) = 0.09$ and $\rho_{AB} = (0.09 - 0.25)/0.25 = -0.64$.

The authors’ federated clients are non-identically distributed by construction, with Dirichlet sampled class counts in their Section 6, and their Figure 5 places a per-client matrix beside one “computed with a global calibration set” and reads the two together. A strong entry of either sign is produced here by the mixture of clients alone. Accuracy is one, so the confusion matrix is diagonal, and there are no A or B inputs for the authors’ reading of the entry to refer to.

Example 7 (Two pairs the model never confuses get different entries). Three classes, cat, dog and plane, with equal frequency. The classifier ranks the true class first on every input and returns the same set for every input of a given class.

True class	Prediction set	z_{cat}	z_{dog}	z_{plane}
cat	{cat, dog}	1	1	0
dog	{dog}	0	1	0
plane	{plane}	0	0	1

A third of the inputs are cats, so $p_{\text{cat}} = 1/3$, and likewise $p_{\text{dog}} = 2/3$ and $p_{\text{plane}} = 1/3$. Each variance is $2/9$. Cat and dog appear together on the cat inputs, so $q_{\text{cat,dog}} = 1/3$. Plane never appears with either, so the other two co-occurrence rates are zero. Substituting,

$$\begin{aligned} \rho_{\text{cat,dog}} &= \frac{1/3 - (1/3)(2/3)}{2/9} = \frac{1}{2}, \\ \rho_{\text{cat,plane}} &= \frac{0 - (1/3)(1/3)}{2/9} = -\frac{1}{2}, \\ \rho_{\text{dog,plane}} &= \frac{0 - (2/3)(1/3)}{2/9} = -1. \end{aligned}$$

The authors say the entries quantify “how easy to confuse those classes are”, and their Remark 2 attaches strong negative entries to dissimilar pairs. Read that way, dog and plane are strongly dissimilar and cat and plane less so. But the classifier treats plane identically with respect to both. It never puts plane in a set with either. The two entries differ only because dog is in twice as many sets as cat. Lemma 2 gives the general statement.

Example 8 (Moving one label occurrence between rows moves an entry from one to almost zero). There are N test inputs, every input has the same true class T , which is in every set and ranked first, and two rare labels A and B each appear in exactly one set. If they appear in the same set then

$z_A = z_B$ and $\rho_{AB} = 1$. If instead the one occurrence of B is in a different set then $p_A = p_B = 1/N$, $q_{AB} = 0$, and

$$\rho_{AB} = \frac{0 - 1/N^2}{(1/N)(1 - 1/N)} = -\frac{1}{N - 1},$$

which at $N = 1000$ is about -0.001 .

The authors present heat maps of entries, their Figures 2 and 5, with no uncertainty attached. For rare labels, moving a single label occurrence from one row to another moves an entry from one to about zero. That is half of the nominal range $[-1, 1]$ and, by Lemma 2, the whole of the range attainable at these inclusion rates, $[-1/(N - 1), 1]$.

4 Two lemmas

Write $o_i = p_i/(1 - p_i)$ for the odds of inclusion of class i .

Lemma 2 (The floor, and the attainable range). *If classes i and j never share a set, then*

$$\rho_{ij} = -\sqrt{o_i o_j} = -\sqrt{\frac{p_i p_j}{(1 - p_i)(1 - p_j)}}. \quad (5)$$

With H classes of equal inclusion rate and singleton sets this equals $-1/(H - 1)$. More generally, for fixed inclusion rates $p_i \leq p_j$ the entry ρ_{ij} ranges over

$$\left[-\sqrt{o_i o_j}, \sqrt{o_i o_j} \right] \quad \text{if } p_i + p_j \leq 1, \quad \left[-1/\sqrt{o_i o_j}, \sqrt{o_i o_j} \right] \quad \text{if } p_i + p_j > 1.$$

Proof. If $q_{ij} = 0$ then (3) reads

$$\rho_{ij} = \frac{-p_i p_j}{\sqrt{p_i(1 - p_i)p_j(1 - p_j)}} = -\sqrt{\frac{p_i^2 p_j^2}{p_i(1 - p_i)p_j(1 - p_j)}} = -\sqrt{\frac{p_i p_j}{(1 - p_i)(1 - p_j)}}.$$

With singleton sets every pair has $q_{ij} = 0$, and with equal inclusion rates $p_i = 1/H$, so $o_i = 1/(H - 1)$ and $\rho_{ij} = -1/(H - 1)$.

For the range, q_{ij} is the probability of the intersection of two events with probabilities $p_i \leq p_j$, so by the Fréchet bounds (Fréchet, 1951) $\max(0, p_i + p_j - 1) \leq q_{ij} \leq p_i$. At the upper bound the numerator of (3) is $p_i - p_i p_j = p_i(1 - p_j)$ and

$$\rho_{ij} = \frac{p_i(1 - p_j)}{\sqrt{p_i(1 - p_i)p_j(1 - p_j)}} = \sqrt{\frac{p_i(1 - p_j)}{(1 - p_i)p_j}} = \sqrt{o_i o_j}.$$

At the lower bound, if $p_i + p_j \leq 1$ then $q_{ij} = 0$ and (5) applies. If $p_i + p_j > 1$ then $q_{ij} = p_i + p_j - 1$, the numerator is $p_i + p_j - 1 - p_i p_j = -(1 - p_i)(1 - p_j)$, and

$$\rho_{ij} = \frac{-(1 - p_i)(1 - p_j)}{\sqrt{p_i(1 - p_i)p_j(1 - p_j)}} = -\sqrt{\frac{(1 - p_i)(1 - p_j)}{p_i p_j}} = -1/\sqrt{o_i o_j}. \quad \square$$

The lemma has two consequences for reading a heat map. The first concerns the scale. An entry can reach $+1$ only when $p_i = p_j$, so the ceiling is below one whenever the two inclusion rates differ, and entries for pairs with different inclusion rates are not on a common scale. This is the classical $\phi/\phi_{\max}$ problem for 2×2 tables, reviewed by Davenport and El-Sanhuray (1991).

The second concerns α . With the model, the calibration data, the auxiliary uniforms and the tie-breaking fixed, the APS sets are nested in α . As α grows no inclusion rate increases, so every floor entry (5) moves toward zero or stays put, although nothing about the classifier has changed, as long as the indicators involved remain nondegenerate. The authors sweep α over $\{0.3, 0.2, 0.1, 0.05\}$ and present the dependence as a feature, their Benefit 3 in Section 5.

Lemma 3 (Identification of the inclusion rates). *Suppose every prediction set has at most one element and at least three classes have inclusion rate strictly between zero and one. Then for any three distinct such classes i, j, l ,*

$$o_i = -\frac{\rho_{ij} \rho_{il}}{\rho_{jl}}, \quad p_i = \frac{o_i}{1 + o_i}. \quad (6)$$

The off-diagonal entries of the matrix therefore determine every inclusion rate exactly.

Proof. With sets of at most one element no two classes ever share a set, so (5) holds for every pair, and $\rho_{ij} \rho_{il} = \sqrt{o_i o_j} \sqrt{o_i o_l} = o_i \sqrt{o_j o_l} = -o_i \rho_{jl}$. Divide by $-\rho_{jl}$, which is nonzero since $o_j, o_l > 0$. The map $o \mapsto o/(1 + o)$ inverts $p \mapsto p/(1 - p)$. $\square$

The authors argue in their Benefit 4 that the matrix is privacy preserving because it “only contains probabilities; hence, it does not leak information on the size and composition of the local datasets”. Correlations can be negative, so the matrix does not contain probabilities. A row-normalized confusion matrix does contain only probabilities, and does not directly disclose counts either.

Lemma 3 says more. When every set has at most one element and at least three classes are nondegenerate, the off-diagonal entries recover every inclusion rate. Inside randomized APS that recovers the class proportions exactly when the classifier is perfect and every class is retained with the same probability, which APS does not guarantee in general but which the next construction arranges.

Example 9 (The matrix reveals the class proportions, within randomized APS). Take a perfect classifier that assigns the ranked probabilities $(0.6, 0.3, 0.1)$ with the true class always first, and let the true class proportions be π_1, π_2, π_3 , all positive. By (1) every calibration score is $0.6 - 0.6u$, so $\tau \in (0, 0.6)$. At a test input the top class has score $0.6 - 0.6U \leq \tau$ exactly when $U \geq 1 - \tau/0.6$, an event of probability $r = \tau/0.6$, and the second has score $0.3 - 0.3U \geq 0.6 > \tau$ and is never included. So the set is $\{Y\}$ with probability r and empty otherwise, every set has at most one element, and $p_i = r \pi_i \in (0, 1)$. Lemma 3 recovers every p_i from the off-diagonal entries, and then

$$\pi_i = \frac{p_i}{\sum_j p_j}.$$

The population matrix determines the true class proportions exactly.

For an empirical matrix the same algebra recovers the empirical inclusion frequencies exactly, provided at least three classes are nondegenerate in the sample and the entries are not rounded, and the proportions are then estimated rather than read off.

The equal top probabilities matter. If class i has top probability a_i and $\tau < \min_i a_i$, then $p_i = \pi_i \tau / a_i$, so the inclusion rates mix the proportions with class-dependent retention. The proportions $(1/3, 1/3, 1/3)$ with top probabilities $(0.6, 0.6, 0.6)$ and the proportions $(5/18, 6/18, 7/18)$ with top probabilities $(0.5, 0.6, 0.7)$ both give, conditional on a common fixed τ below 0.5, $p_i = 5\tau/9$ for every class, hence the same law of sets and the same matrix.

The matrix is offered as useful because it exposes how a client’s data differs from the others. No threat model, mechanism or leakage bound is given. Inferring a client’s label distribution is a recognized attack in federated learning, from gradients (Gu and Bai, 2023) or from model behaviour over rounds (Ma et al., 2025). In the setting of Example 9 the matrix hands the same quantity over directly.

5 What the coverage guarantee constrains

The authors state in their Section 3.2 that “for a given input sample x CP will construct a random a prediction set $\mathcal{C}(x)$ such that it is guaranteed to contain the correct class lies in with a probability close to $1 - \alpha$ ”, quoted with its typographical errors. The guarantee (4) is marginal. It does not imply coverage conditional on the input, on a class or on a client. In distribution-free regression, exact pointwise conditional coverage forces prediction intervals of infinite expected length at almost every nonatomic point (Barber et al., 2021, Proposition 1, restated from Vovk 2012 and Lei and Wasserman 2014). The client and class example below makes the classification point directly.

The distinction matters here because the authors’ own statement of the guarantee is per input, and the per-client and per-pair matrices of their Sections 5 and 6 inherit that reading with no per-client or per-class coverage reported. Validity does not transfer that way. Take ten equally weighted clients, nine with coverage one and one with coverage zero. Pooled coverage is exactly 0.9, which meets a nominal $1 - \alpha = 0.9$, and the tenth client has no protection at all. The same arithmetic applies to a class with a tenth of the data that is missed every time.

The authors also write the upper coverage term as $1/(n - 1)$, with n their calibration size. For the randomized procedure with almost surely distinct scores the term is $1/(n + 1)$ (Romano et al., 2020, Theorem 1), and Theorem 2.2 of Lei et al. (2018) gives the same bound written as $2/(n + 2)$ with n the total sample, so theirs is looser than necessary and probably a typographical error.

The federated experiments compare local matrices with “CCM entries computed with a global calibration set”, the caption of their Figure 5, across clients that are non-identically distributed by construction. If that global threshold is applied to a client’s local test data, exchangeability fails and even the marginal guarantee need not hold. No coverage or set sizes are reported for any client.

6 What to report instead

If the question is which classes the classifier confuses, the object that answers it is the confusion matrix of the ranked prediction, $\mathbb{P}(\hat{Y} = b \mid Y = a)$, which needs labeled evaluation data. Earlier sections used the joint form $\mathbb{P}(Y = a, \hat{Y} = b)$. The two differ by the class proportion, which every construction in this note preserves, so the results hold for either. A complementary summary of how the classifier allocates probability is the soft confusion matrix

$$M_{ab} = \mathbb{E}[f_b(X) \mid Y = a].$$

This is the average probability assigned to class b on inputs of true class a . It is directional and threshold-free, it needs no calibration set, and like any confusion matrix it needs labeled evaluation data.

The matrix M_{ab} is a conditional mean, so it summarizes rather than determines. With two classes, a model with $f_B(X) = 0.4$ on every input of class A and a model with $f_B(X)$ equal to 0.6 or 0.2 with equal probability have the same row (0.6, 0.4), and error rates from A to B of zero and one half. The confusion matrix of the ranked prediction carries that difference and M_{ab} does not, which is why the two are complements.

If labels are unavailable, the mean pairwise product $\mathbb{E}[f_i(X)f_j(X)]$ is symmetric and needs no labeled calibration or evaluation sample. If sets are the object, because a downstream consumer receives sets and nothing else, then the co-occurrence rate q_{ij} is the quantity the abstract describes, and the coverage confusion matrix of Stutz et al. (2022), which stratifies by the true label, is directional. Divided by the true class frequency it gives the conditional version $\mathbb{P}(b \in C(X) \mid Y = a)$. The row sum is the class frequency times the expected set size within the class, since one input can contribute to several columns, so dividing by the row sum instead divides the conditional version by that expected set size.

In every case the inclusion rates and the set sizes should be reported alongside, since by Lemma 2 an entry of (3) cannot be read without them.

The authors cite the coverage confusion matrix in their related work and compare against none of these. Their only comparator, in their Benefits 1, 2 and 4, is the hard confusion matrix, which discards the probabilities entirely, so the contrast they draw is soft versus hard rather than conformal versus not.

The authors’ client selection experiment does not bear on any of this, but we note five things. It selects the clients with the highest entries for two critical classes, while the centralized section reads a falling entry as the classifier learning to separate them. It generates heterogeneity with a symmetric Dirichlet($\beta, \dots, \beta$) and states that larger β gives more heterogeneous clients, whereas $\text{Var}(\theta_h) = (H - 1)/(H^2(H\beta + 1))$ decreases in β . Its Figures 6 and 7 are captioned as matrix entries while their axes are class accuracy, and its Table 1 lists six selected CIFAR-10 clients where the text says seven. The same table reports accuracy on the noncritical classes moving from 0.513 to 0.494 on CIFAR-100 and from 0.859 to 0.856 on CIFAR-10, and overall CIFAR-100 accuracy from 0.511 to 0.493, which the authors summarize first as “small effects on the accuracy of non-critical classes” and then as “does not affect the performance for non-critical classes”, without repetitions, seeds or error bars.

Simulations of Example 1 on its three classes, and, on a synthetic ten-class classifier with adaptive prediction sets, of Examples 5 and 7 and of the direction blindness of Example 2, with the soft confusion matrix and the mean pairwise product alongside as α varies, are available at <https://conformalprediction.net/demos/32-conformal-correlation.html>. A script that re-computes every quantity in this note, exactly where the arithmetic is rational and by simulation for the claims about randomized APS, the Beta law of the realized coverage and the Dirichlet variance, is at <https://github.com/microprediction/conformalprediction/tree/main/papers/conformal-correlation-counterexamples>.

7 Conclusion

The conformal correlation matrix is the phi coefficient of APS set membership indicators. By Observation 1 it sees only the law of the sets, by Lemma 1, with three or more classes, the marginal coverage lower bound leaves an off-diagonal entry free among the laws that satisfy it, and by Theorem 1, over joint laws constrained only by $\hat{Y} \in C(X)$ whenever $C(X) \neq \emptyset$, the law of the sets does not identify the confusion matrix once a set of size at least two occurs, with Example 1 exhibiting the corresponding failure inside randomized APS.

Example 1 shows, inside randomized APS itself, that the matrix does not determine the confusion matrix. Examples 2 through 8 concern laws of sets that meet the coverage bound. Among them, Example 3 disproves the converse of the authors’ Remark 3, Example 4 shows that an entry does not determine the co-occurrence probability, and Examples 5 and 6 show positive entries with no pairwise confusion. Lemma 2 shows that for pairs that never share a set the entry is a function of

the inclusion rates alone, and Lemma 3 with Example 9 shows that inside randomized APS a perfect classifier’s matrix reveals the class proportions. None of this is repaired by the coverage guarantee, which concerns the true label and does not generally determine which other labels sit beside it.

References

- R. F. Barber, E. J. Candès, A. Ramdas, and R. J. Tibshirani. The limits of distribution-free conditional predictive inference. *Information and Inference*, 10(2):455–482, 2021.
- P. Cotton. Marginally useful: an information-gap identity in conformal prediction. arXiv preprint arXiv:2608.07479v2, 2026.
- E. C. Davenport and N. A. El-Sanhury. Phi/phimax: Review and synthesis. *Educational and Psychological Measurement*, 51(4):821–828, 1991.
- M. Fréchet. Sur les tableaux de corrélation dont les marges sont données. *Annales de l’Université de Lyon, Section A*, 14:53–77, 1951.
- E. Garcia-Ceja, L. Garcia-Banuelos, and N. Jourdan. Conformal prediction in multi-user settings: An evaluation. arXiv:2312.05195, 2023.
- Y. Gu and Y. Bai. LDIA: Label distribution inference attack against federated learning in edge computing. *Journal of Information Security and Applications*, 74:103475, 2023.
- J. Lei, M. G’Sell, A. Rinaldo, R. J. Tibshirani, and L. Wasserman. Distribution-free predictive inference for regression. *Journal of the American Statistical Association*, 113(523):1094–1111, 2018.
- Z. Ma, H. Gao, J. Huang, and P. Wang. Whispers of data: Unveiling label distributions in federated learning through virtual client simulation. arXiv:2504.21436, 2025.
- A. Perlo, C. F. Chiasserini, G. De Veciana, and F. Malandrino. Characterizing the performance of classification models through conformal correlation matrices. *Computer Communications*, 247:108398, 2026.
- Y. Romano, M. Sesia, and E. J. Candès. Classification with valid and adaptive coverage. In *Advances in Neural Information Processing Systems*, volume 33, pages 3581–3591, 2020.
- D. Stutz, K. Dvijotham, A. T. Cemgil, and A. Doucet. Learning optimal conformal classifiers. In *International Conference on Learning Representations*, 2022.
- V. Vovk, A. Gammerman, and G. Shafer. *Algorithmic Learning in a Random World*. Springer, 2005.
- G. U. Yule. On the methods of measuring association between two attributes. *Journal of the Royal Statistical Society*, 75(6):579–652, 1912.